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MILITARY SPECIFICATION

ACTUATOR, PISTON, PA535
PARTS FOR AND
LOADING, ASSEMBLING AND PACKING

This specification is approved for use by the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements and quality assurance provisions for the fabrication of parts, assembling, and packing for one type of piston actuator designated as Actuator, Piston, PA535.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document, should be addressed to: Commander, U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1375

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SPECIFICATIONS

FEDERAL

A-A-208 - Ink Marking, Stencil Opaque (Porous and Non-Porous Surfaces)

MILITARY

MIL-P-116 - Preservation, Methods Of
MIL-B-117 - Bag, Sleeve and Tubing - Interior Packing
MIL-G-45204 - Gold Plating, Electrodeposited
MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification For

STANDARDS

MILITARY

MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-1169 - Packaging, Packing and Marking for Shipment of Inert Ammunition Components
MIL-STD-1190 - Minimum Guidelines for Level C Preservation Packing and Marking

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from DODSSP - Customer Service, Standardization Documents Order Desk, 700 Robbins Avenue, Bldg. 4D, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS (see 6.5)

U.S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

PRODUCT AND PACKAGING DRAWINGS

8796522 - Marking Diagram and Sealing for Wood Packing Boxes
12550906 - Actuator, Piston, PA535
12550757 - Ballistic Environment Test Requirement of M762/M767 Components and Subassemblies

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12631059 - Actuator, Piston, PA535 Test Fixture Assembly

(Copies of other Government documents, drawings, and publications, required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

PUBLICATIONS

CODE OF FEDERAL REGULATIONS

Title 49 - Transportation, Parts 100-199

(The Code of Federal Regulations is available from the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402. Orders should specify, "49 CFR 100-177 (latest revisions)".)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained. (See contract provisions for additional precedence criteria).

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with the applicable drawings and specifications.

3.2 Components and assemblies. The components and assemblies shall comply with all requirements specified on Drawing 12550906, all associated drawings, and with all requirements specified in applicable specifications and standards.

3.3 Functioning tests - sensitivity and output.

3.3.1 No-fire test. The piston actuator, after being conditioned at 70 degrees ± 10 degrees F for two hours minimum, shall not exhibit any signs of the output requirements of 3.3.2 after being exposed to the no-fire stimulus specified in 4.5.1.1.

3.3.2 All-fire output test. The piston actuator, after being conditioned at 70 degrees ± 10 degrees F for two hours minimum, shall function on the first attempt, shall shear a .047 ± .001 inch diameter, aluminum pin, and travel a minimum of .200 inch when tested as specified in 4.5.1.2 (See Dwg. 12631059).

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3.3.3 No load output. The piston shall move out of the cup when fired in an uncontained mode. The header shall remain intact.

3.4 Environmental conditioning.

3.4.1 Resistance between electrode and cup, thermal shock, vibration and function. The piston actuator shall comply with the resistance requirement of Drawing 12550906, with the output requirements of 3.3.2 and shall function on the first attempt after being exposed to thermal shock and vibration when tested as specified in 4.5.2.1.

3.4.2 Waterproofness. The actuator shall comply with the output requirements of 3.3.2 after being submerged in water to a minimum depth of 2 inches for not less than 48 hours, at a temperature of 70 degrees ± 10 degrees F when tested as specified in 4.5.2.2.

3.4.3 Cold temperature. The piston actuator assembly shall comply with the output requirements of 3.3.2 after being conditioned at minus 65 degrees ± 5 degrees F for two hours minimum and tested as specified in 4.5.2.3.

3.4.4 Hot temperature. The piston actuator assembly shall comply with the output requirements of 3.3.2 after being conditioned at 160 degrees ± 5 degrees F for two hours minimum and tested as specified in 4.5.2.4.

3.4.5 Shock G. The piston actuator assembly shall comply with the output requirements of 3.3.2 after being subjected to a 30000g setback force as specified in 4.5.2.5. The piston actuator assembly shall be conditioned at 70 degrees ± 10 degrees F for 2 hours minimum prior to testing.

3.5 First article. When specified in the contract or purchases order (see 6.2), a sample shall be subjected to first article inspection in accordance with the technical provisions herein (see 4.3).

3.6 Change of qualified product. After first article, the manufacturer shall notify the purchaser prior to the implementation of any change which may affect performance, quality, reliability or interchangeability. Such notification shall include a thorough description of the proposed change and a suggested test plan designed to demonstrate that the change will not adversely affect performance, quality, reliability or interchangeability and that the changed product will continue to meet the specification requirement. The purchaser will review the proposed change and

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test plan and advise the manufacturer of the testing required to obtain approval of the proposed change. Upon completion of the prescribed tests, the results shall be submitted to the purchaser for review. Based upon the review the purchaser will approve or disapprove the change and notify the manufacturer. At the manufacturer's option, devices incorporating the change may be manufactured and tested prior to approval however, all shipments of these changed devices shall be withheld until approval is granted by the purchaser. Changes representative of those which are subject to this requirement are any design changes in the manufacturing processing such as:

a. Package structure and material changes or changes to the geometry.

b. Changes to process flow charts, assembly flow chart and testing flow charts.

c. Conductor materials.

d. Assembly operations move from one line or building to another.

3.7 Workmanship. All parts and assemblies shall be fabricated and assembled in a thorough, workmanlike manner and all manufacturing, processing, assembling operations shall be correctly performed. All parts and assemblies shall be clean, free of burrs, sharp edges, cracks, dirt, grease, rust and other foreign matter that may impair proper functioning. The cleaning method used shall not be injurious to any parts, nor shall the parts be contaminated by the cleaning agents. All required markings and stampings shall be neat and sharply defined.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items shall meet all requirements of Sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 General provisions. Unless otherwise specified herein, the provisions of MIL-A-48078 apply and form a part of this specification. Reference shall be made to MIL-STD-109 to define quality assurance terms used herein.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the assemblies, components and test specimens listed below in the quantities indicated.

PART DESCRIPTION	DRAWING	QUANTITY
Cup	12631048	25
Bridge Plug and Ignition Assembly	12550782	25
Plug Assembly	12550784	25
Header	12631057	25
Piston	12631050	25
Actuator, Piston, PA535	12550906	250

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4.3.2 Inspection to be performed. See MIL-A-48078 and TABLE I as specified herein.

4.3.3 Rejection. See MIL-A-48078 and TABLE I as specified herein.

4.3.4 Component qualification. Qualification of the component supplied to this specification is not complete until it has been submitted to and passed additional tests performed by the purchaser. These tests may include 30,000 G acceleration tests, 30,000 RPM spin tests and aging tests to evaluate long term storage characteristics. Coordination of testing and test reporting shall be the responsibility of the purchaser.

4.3.4.1 Re-qualification. Re-qualification of changed component(s) may depend on the results of circuit and application analysis conducted as a result of change(s).

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TABLE I. First article inspection

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
	PA535 Piston Actuator and Components	1 OF 2	See Below
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
	Actuator, Piston, PA535 (Dwg. 12550906)			
	Examination for defects	250 Note 1	3.2	4.4.2.6
	Functioning tests-sensitivity and output			
	a) No-fire test	0-1 Note 1	3.3.1	4.5.1.1
	b) All-fire output test	75-0-1 Note 2	3.3.2	4.5.1.2
	Environmental tests			
	a) Resistance between electrode & cup, thermal shock, vibration, & function	50-0-1 Note 2	3.4.1	4.5.2.1
	b) Waterproofness test	25-0-1 Note 2	3.4.2	4.5.2.2
	c) Cold test	25-0-1 Note 2	3.4.3	4.5.2.3
	d) Hot test	25-0-1 Note 2	3.4.4	4.5.2.4
	e) Shock, G Test	25-0-1 Note 2	3.4.5	4.5.2.5
	No Load Test	25-0-1	3.3.3	4.5.1.3

NOTES:
1. The entire quantity of piston actuators shall be subjected to the specific examination/test prior to testing. 2. Samples for this test shall be taken from no-fire test samples.

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TABLE I. First article inspection

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
	PA535 Piston Actuator and Components	2 OF 2	See Below

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	NEXT HIGHER ASSEMBLY / INSPECTION METHOD REFERENCE
	Cup (Dwg. 12631048)			
	Examination for defects	25-0-1	3.2	4.4.2.1
	Bridge Plug and Ignition Assembly (Dwg. 12550782)			
	Examination for defects	25-0-1	3.2	4.4.2.2
	Plug Assembly (Dwg. 12550784)			
	Examination for defects	25-0-1	3.2	4.4.2.3
	Insulation resistance between electrode and header of plug assembly	25-0-1	3.2	4.5.4
	Pull test of electrode	25-0-1	3.2	4.5.7
	Header (Dwg. 12631057)			
	Examination for defects	25-0-1	3.2	4.4.2.4
	Piston (Dwg. 12631050)			
	Examination for defects	25-0-1	3.2	4.4.2.5

NOTES:

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, inspection lots shall contain parts bearing the same lot interfix number from one manufacturer.

4.4.2 Examination and tests.

a. Classification of characteristics. Quality conformance examinations and tests are specified in the following Classification of Characteristics paragraphs. The contractor's quality program or detailed inspection system shall provide assurance of compliance of all characteristics with the applicable drawing and specification requirements utilizing as a minimum the conformance criteria specified. When cited herein, attributes sampling inspection shall be conducted in accordance with TABLE II below, using the inspection levels stated in the Classification of Characteristics paragraphs.

TABLE II Attributes sampling inspection

	Inspection Levels					
Lot Size	I	II	III	IV	V	VI
2 to 8	*	*	*	*	5	3
9 to 15	*	*	*	13	5	3
16 to 25	*	*	*	13	5	3
26 to 50	*	*	32	13	5	3
51 to 90	*	*	32	13	13	5
91 to 150	*	125	32	13	13	5
151 to 280	*	125	32	32	20	8
281 to 500	*	125	32	32	20	8
501 to 1200	*	125	80	50	20	13
1201 to 3200	1250	125	80	50	32	13
3201 to 10000	1250	125	125	50	32	13
10001 to 35000	1250	315	125	80	50	13
35001 to 150000	1250	315	125	80	50	13
150001 to 500000	1250	500	200	125	50	13
500001 and over	1250	500	200	125	50	13

Numbers under inspection levels indicate sample size; asterisks (*) indicate one hundred percent inspection. If sample size exceeds lot size, perform one hundred percent inspection. Accept on zero and reject on one or more failures/defects for all inspection levels.

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b. Alternative quality conformance provisions. Unless otherwise specified herein or provided for in the contract, alternative quality conformance procedures, methods or equipment, such as statistical process control, tool control, other types of sampling plans, etc., may be used by the contractor when they provide, as a minimum, the level of quality assurance required by the provisions herein. Prior to applying such alternative procedures, methods or equipment, the contractor shall describe them in a written proposal submitted to the Government for evaluation (see 6.6). When required, the contractor shall demonstrate that the effectiveness of each proposed alternative is equal to or better than the specified quality conformance provision(s) herein. In case of dispute as to whether the contractor's proposed alternative(s) provides equivalent assurance, the provisions of this specification shall apply. All approved alternative provisions shall be specifically incorporated into the contractor's quality program or inspection system, as applicable.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Cup	1 OF 1	12631048
			NEXT HIGHER ASSEMBLY: 12550906

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Inside diameter, min.	Level IV	3.2	Gage
102	True position of electrode hole incorrect	Level IV	3.2	Gage
103	Overall length	Level IV	3.2	Gage
104	Cup dented, distorted, or otherwise damaged	Level IV	3.2	Visual
Minor				
201	Outside diameter (O.D.)	Level V	3.2	Gage
202	Evidence of Poor Workmanship	Level V	3.7	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.2	Bridged Plug and Ignition Assembly	1 OF 1	12550782
			NEXT HIGHER ASSEMBLY: 12550906

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Spot charge missing	Level IV	3.2	Visual
102	Cracks or voids in spot charge	Level IV	3.2	Visual
104	Ignition material on outside of plug	100%	3.2	Visual
Minor				
201	Evidence of Poor workmanship	Level V	3.7	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.3	Plug Assembly	1 OF 1	12550784
			NEXT HIGHER ASSEMBLY: 12550783

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Pull test of electrode	50-0-1	3.2	4.5.7
102	Insulation resistance test	100%	3.2	4.5.4
103	Outside diameter	Level IV	3.2	Gage
104	Plating improper	Note 1	3.2	4.5.6
105	Glass pellet cracked	Level IV	3.2	Visual
106	Length of exposed pellet, max	Level IV	3.2	Gage
Minor				
201	Evidence of Poor workmanship	Level V	3.7	Visual

NOTES:
1. Sampling and rejection shall be in accordance with MIL-G-45204.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.4	Header	1 OF 1	12631057
			NEXT HIGHER ASSEMBLY: 12550784

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Small inside diameter	Level IV	3.2	Gage
102	True position of large I.D.	Level IV	3.2	Gage
103	True position of small outside diameter	Level IV	3.2	Gage
104	Total length	Level IV	3.2	Gage
105	Length of small O.D.	Level IV	3.2	Gage
Minor				
201	Large O.D.	Level V	3.2	Gage
202	Samll O.D.	Level V	3.2	Gage
203	Evidence of Poor workmanship	Level V	3.7	Visual

NOTES:

AMSMC Form 1570b-E, 1 Jul 89 Replaces AMSMC Form 1570, 1 Feb 85, which may not be used.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.5	Piston	1 OF 1	12631050
			NEXT HIGHER ASSEMBLY: 12631049

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Total length	Level IV	3.2	Gage
102	Length of cavity	Level IV	3.2	Gage
103	Outside diameter	Level IV	3.2	Gage
104	Inside diameter	Level IV	3.2	Gage
105	Surface finish of outside diameter improper	Level IV	3.2	Gage
Minor				
201	True position of inside diameter	Level V	3.2	Gage
202	Evidence of poor workmanship	Level V	3.7	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.6	Actuator, Piston, PA535	1 OF 1	12550906
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Special				
a	Cup cracked or split	100%	3.2	Visual
b	Explosive on exterior of cup	100%	3.2	Visual
Major				
101	Length of electrode	Level IV	3.2	Gage
102	Length, sealed end to base	Level IV	3.2	Gage
103	Diameter of cup	Level IV	3.2	Gage
104	Resistance between electrode and cup	Level IV	3.2	Gage, 4.5.5
105	Sealant not present 360 degrees	100%	3.2	Visual
106	Presence of sealant on electrode beyond sealed area	Level IV	3.2	Visual
107	Sensitivity - no-fire	200-0-1	3.3.1	4.5.1.1
108	Functioning - all-fire	50-0-1 Notes 1&2	3.3.2	4.5.1.2
109	Environmental test			
	a) Resistance between electrode and cup, thermal shock, vibration and function	50-0-1 Note 2	3.4.1	4.5.2.1
	b) Waterproofness	25-0-1 Note 2	3.4.2	4.5.2.2
	c) Cold temperature	25-0-1 Note 2	3.4.3	4.5.2.3
	d) Hot temperature conditioning	25-0-1 Note 2	3.4.4	4.5.2.4
110	No. load test	25-0-1 Note 2	3.3.3	4.5.1.3
Minor				
201	Evidence of poor workmanship	Level V	3.7	Visual

NOTES:
1. Any actuator that fails to function on first attempt or fails to produce the required output shall be classed defective. 2. Samples from this test shall be taken from no-fire samples.

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4.4.3 Testing. Testing is described in the First Article Inspection and Quality Conformance Inspection tables.

4.4.4 Inspection equipment. The inspection equipment required to perform the inspections specified herein is identified in the "Inspection Method Reference" column of the Classification of Characteristics listings starting with 4.4.2.1. Contractor inspection equipment designs shall be submitted for Government approval as specified in the contract. See Section 6 of MIL-A-48078 and 6.3 herein.

4.5 Methods of inspection.

4.5.1 Functioning tests - sensitivity and output.

4.5.1.1 No-fire test. The piston actuator, maintained at the conditioning temperature, shall be subjected to the specified no-fire stimulus in accordance with the requirements of Drawing 12550906. Any sample which fails to comply with the requirement of 3.3.1 shall be classed defective and shall cause the lot to be rejected. (Non-destructive test).

4.5.1.2 All-fire output. The piston actuator, maintained at the conditioning temperature, shall be assembled in the specified test fixture, also maintained at the conditioning temperature, (Dwg. 12631059) and subjected to the specified all-fire stimulus in accordance with the requirements of Drawing 12550906. Observation shall be made for compliance with the requirements of 3.3.2. Any sample which fails to fire on the first attempt shall be classed defective. (Destructive test).

4.5.1.3 No load output. The actuator shall be positioned unconstrained in a test fixture and discharged. The piston shall leave the cup and the header remain in the cup.

4.5.2 Environmental tests.

4.5.2.1 Resistance between electrode and cup, thermal shock, vibration, and function. The piston actuators shall be tested as follows, in the sequence specified, using approved equipment:

a. Measure resistance between the electrode and cup in accordance with 4.5.5.

b. Condition for two (2) hours minimum at minus 65 degrees ± 5 degrees F.

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c. Remove from cold chamber and within 20 seconds transfer to a hot chamber at 160 degrees ± 5 degrees F.

d. Condition for two (2) hours minimum at 160 degrees ± 5 degrees F and 20% maximum relative humidity.

e. Remove from hot chamber and immediately transfer to the vibrator.

f. Vibrate for 15 minutes, minimum, in a sweep covering a frequency range of 25-500-25 cycles per second, parallel to the transverse orthogonal axis, with output end down and a 2 ± 0.2 G's peak amplitude.

g. Measure resistance between electrode and cup in accordance with 4.5.5.

h. The piston actuators shall be tested as specified 4.5.1.2. Any piston actuator which fails to comply with the applicable requirement shall be classed defective. (Destructive test).

4.5.2.2 Waterproofness. The piston actuators shall be immersed in water at a two (2) to three (3) inch depth for 48 hours, minimum at the conditioning temperature. At the end of the immersion time, the piston actuators shall be removed from the water, tested according to 4.5.1.2, and observed for compliance with the requirements of 3.3.2. Any sample which fails to comply with the applicable requirements shall be classed defective. (Destructive test).

4.5.2.3 Cold temperature. The piston actuator assemblies, maintained at the conditioning temperature, shall be functioned as specified in 4.5.1.2 (Destructive test).

4.5.2.4 Hot temperature. The piston actuator assemblies, maintained at the conditioning temperature, shall be functioned as specified in 4.5.1.2 (Destructive test).

4.5.2.5 Shock, G. The piston actuator assemblies, maintained at the conditioning temperature, shall be subjected to a 30,000g setback force in accordance with the ballistic environment test requirements of M762/M767 components and subassemblies, 12550757. After completion of the test, the piston actuator assemblies shall be tested as specified in 4.5.5 (destructive test).

4.5.3 Resistance between electrode and header. The bridge assembly shall be connected to the equipment and the bridge

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resistance shall be measured when a current limited to one (1) milliampere is applied. Any bridge assembly that fails to comply with the drawing requirement shall be classed defective and removed from the lot. (Non-destructive test).

4.5.4 Insulation resistance between electrode and header of plug assembly. The required D.C. voltage shall be applied to the plug assembly between the electrode and the header for not less than one or more than two seconds. If the resistance measures less than the specified resistance, the plug assembly shall be classed defective and removed from the lot. (Non-destructive test).

4.5.5 Resistance between electrode and cup. The piston actuator assembly shall be connected to the equipment and the resistance shall be measured between the electrode and the cup when a current limited to ten (10) milliampere is applied. Any actuator assembly that fails to comply with the drawing requirement shall be classed defective and removed from the lot. (Non-destructive test).

4.5.6 Plating of electrodes. The plating shall be verified with the applicable drawing requirement. (Test samples shall not be returned to the lot).

4.5.7 Pull test of electrode of plug assembly. The plug assembly shall be tested in accordance with the applicable drawing requirements. (Destructive test).

5. PACKAGING

5.1 Unit Packaging

5.1.1 Level A. Not Applicable.

5.1.2 Level B. Not Applicable.

5.1.3 Level C. Actuators, Piston shall be packaged in accordance with MIL-STD-1190 and the applicable section (s) of Code of Federal Regulations, Title 49. In addition, each piston actuator shall be electrically shorted during shipment and storage by means of a metallic connection between electrode and cup.

Unless other packaging is used to provide barrier protection, the following instructions apply:

Each intermediate fiberboard box, set-up box or comparable container, after blunting corners, shall be placed in a water and vapor proof bag manufactured in accordance with Type I, Class E of

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MIL-B-117. The bag shall be closed. All excess air shall be exhausted from within the bag and the bag then sealed in accordance with the manufacturer's instructions. The seal shall be located at the edge of the bag to provide for a minimum of two (2) subsequent seals, if required. The seal shall comply with the heat seal requirements of MIL-P-116.

5.2 Packing.

5.2.1 Level A. Not Applicable.

5.2.2 Level B. Not Applicable.

5.2.3 Level C. A suitable number of actuators piston packaged as specified in 5.1.3, shall be packed in an outer box in accordance with MIL-STD-1190 and the applicable section(s) of Code of Federal Regulations, Title 49.

5.3 Marking.

5.3.1 Level A. Not Applicable.

5.3.2 Level B. Not Applicable.

5.3.3 Level C. Each inner container and intermediate fiberboard box, set-up box or comparable container shall be printed, stamped or labeled and each barrier bag printed or stamped as required by the applicable section(s) of Code of Federal Regulations, Title 49, and in accordance with MIL-STD-129, and the following information:

a. (Number) Actuator, Piston, PA535: Lot Number.

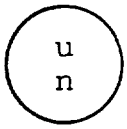
b. All letters shall be upper case. Size of letters and numerals shall be a minimum of 1/4" high applied with black waterproof ink. Stencil ink shall conform to A-A -208.

c. Printing and labeling shall be commercial type. Labels shall be on white gummed paper.

d. Exterior boxes shall be marked in accordance with Drawing 8796522. The DOT designation shall be in accordance with U.S. Army ARDEC Hazard Classification Safety Data statement. Apply appropriate label.

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5.3.3.1 Performance oriented packaging marking. Apply POP markings on side of box opposite nomenclature:



4C1/Y*/S/**
USA/DOD/AYD

Note 1 - *1/2 the weight of pack in pounds, plus 1.
Note 2 - ** Date packaged.

5.3.3.2 Unitization. Unitization shall be in accordance with MIL-STD-1190.

5.4 Packaging, packing and marking for inert piston actuator parts. Packaging, packing and marking for inert piston actuator parts shall be in accordance with MIL-STD-1169 for all applicable levels of packaging.

5.5 Shipping. When piston actuators from more than one lot are shipped at one time, each lot shall be kept separate and the division between lots clearly indicated to prevent mixing of lots in transit.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The Actuator, Piston, PA535 is intended for use in munition assemblies.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and, if required, the specific issue of individual documents referenced (see 2.1.1)

c. Requirements for submission of first article sample.

6.3 Submission of contractor inspection equipment designs for approval. Submit copies of designs as required to: Commander, U.S. Army ARDEC, ATTN: SMCAR-QAR-I, Picatinny Arsenal, New Jersey 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract.

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6.4 Distribution of ammunition data cards, inspection and test results, process control. Distribution of ammunition data cards, inspection and test process control shall be sent to:

Commander
ARDEC
ATTN: SMCAR-QAR-I
Picatinny Arsenal, New Jersey 07806-5000.

6.5 Drawings. Drawings listed in Section 2 of this specification under the heading U.S. Army Armament Research, Development, and Engineering Center (ARDEC) may also include drawings prepared by, and identified as U.S Army Armament, Research and Development Command (ARRADCOM), Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARDEC.

6.6 Submission of alternative quality conformance provisions. All contractor proposed alternative quality conformance provisions will be submitted to the Government for evaluation/approval as directed by the contracting activity.

6.7 Subject term (key word) listing.

Shock G Test
Waterproofness

6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1375-0007)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-F-70827A (AR)	940602

3. DOCUMENT TITLE ACTUATOR, PISTON, PA535 PARTS FOR AND LOADING, ASSEMBLING AND PACKING

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets if needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
U.S. ARMY ARDEC STANDARDIZATION OFFICE	(1) Commercial 201-724-6675	(2) AUTOVON DSN-880-6675
c. ADDRESS (Include Zip Code) ATTN: SMCAR-BAC-S PICATINNY ARSENAL, NJ 07806-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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